The axle lock switch in the accessory switch bezel on the instrument panel provides a hard wired ground input to the instrument cluster circuitry through the front axle lock request circuit whenever the switch is enabled and the rear and front axle lock position of the switch is selected by the vehicle operator. The front locker indicator switch on the front axle housing provides a hard wired ground input to the instrument cluster whenever the front axle locker mechanism is engaged. The key-in ignition switch integral to the ignition switch provides a hard wired ground input to the instrument cluster whenever a key is present in the ignition lock cylinder. Each of these inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the instrument cluster circuitry that controls the front lock indicator LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

FUEL GAUGE

DESCRIPTION

Fig. 15 Fuel Gauge Icon

A fuel gauge is standard equipment on all instrument clusters. The fuel gauge is located in the lower left quadrant of the instrument cluster, below the voltage gauge. The fuel gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from “E” (or Empty) to “F” (or Full). An International Control and Display Symbol icon for “Fuel” is located on the cluster overlay, in the center of the gauge directly above the hub of the gauge needle (Fig. 15). An arrowhead pointed toward the left side of the vehicle is imprinted on the cluster overlay next to the “Fuel” icon in the fuel gauge to provide the driver with a reminder as to the location of the fuel filler access. The fuel gauge graphics are white and blue against a black field except for a single red graduation at the low end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The fuel gauge is serviced as a unit with the instrument cluster.

OPERATION

The fuel gauge gives an indication to the vehicle operator of the level of fuel in the fuel tank. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The fuel gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full, the cluster programming applies an algorithm to calculate the proper gauge needle position, then moves the gauge needle to the proper relative position on the gauge scale. The algorithm is used to dampen gauge needle movement against the negative effect that fuel sloshing within the fuel tank can have on accurate inputs from the fuel tank sending unit to the PCM.
- **Less Than 12.5 Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than about 12.5 (one-eighth), the gauge needle is moved to the proper position on the gauge scale, the low fuel indicator is illuminated, and a single chime tone is sounded. The low fuel indicator remains illuminated until the cluster receives messages from the PCM for a continuous twenty seconds indicating that the percent tank full has increased by more than 0.625 gallons or until the ignition switch is turned to the Off position, whichever occurs first. The chime tone feature will occur only once per ignition cycle.
- **Less Than Empty Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than empty, the gauge needle is moved to the far left end of the gauge scale and the low fuel indicator is illuminated immediately. This message would indicate that the fuel tank sender input to the PCM is a short circuit.
- **More Than Full Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full is more than full, the gauge needle is moved to the far left end of the gauge scale and the low fuel indicator is illuminated immediately. This message would indicate that